Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 24
Pseudo-Inverses and Least Squares.

Review

1. Show: if A is a matrix with linearly independent columns then A^TA is invertible.
2. We showed: if A is a matrix with linearly independent columns then $(A^TA)^{-1}A^T$ is the left-inverse of A (called a *pseudo-inverse*). Call it $A^\dagger$.

Least Squares Approximations

1. Argue: for any vectors $\vec{a}$ and $\vec{b}$,

$$||\vec{a}+\vec{b}||^2 = ||\vec{a}||^2 + ||\vec{b}||^2 + 2\vec{a}^T\vec{b}.$$

2. Suppose A has independent columns, and thus A^TA has an inverse.

   Prove: $\tilde{x} = (A^TA)^{-1}A^T\vec{b}$ is the "best" approximate solution to $A\vec{x} = \vec{b}$ (in the sense that $||A\tilde{x} - \vec{b}|| \leq ||A\vec{x} - \vec{b}||$ for every vector $\vec{x}$).

   Data: Points $(1,1), (2,3), (3,2)$ in $\mathbb{R}^2$.

   Goal: Find the least-squares best fit line for this data. Assume our model is *linear* of the form $y = b + mx$, where b is the y-intercept of the line and m is the slope. Ideally we'd want each point (x_i, y_i) to satisfy:

$$y_i = b + mx_i,$$

3. Write out the desired equations.

4. Argue that no linear function can exactly satisfy this data.

5. Write these equations in matrix form.

6. Let $\tilde{x}$ be the least squares best fit solution (given by the pseudo-inverse formula). Find $\tilde{x}$.

7. What is the linear function? Use it to find an estimated y when $x = 3$.

   Data: Points $(-1, 1), (0, 0), (1, 2)$ in $\mathbb{R}^2$.

   Goal: Find the least-squares best fit line for this data. Assume our model is *linear* of the form $y = b + mx$, where b is the y-intercept of the line and m is the slope. Ideally we'd want each point (x_i, y_i) to satisfy:

   $$y_i = b + mx_i,$$

8. Write out the desired equations.

9. Argue that no linear function can exactly satisfy this data.

10. Write these equations in matrix form.

11. Let $\tilde{x}$ be the least squares best fit solution (given by the pseudo-inverse formula). Find $\tilde{x}$.

12. What is the linear function? Use it to find an estimated y when $x = 3$.